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Dear Editor,

We are pleased to submit our manuscript entitled “Design and Evaluation of an Open-Source LLM-Based Intelligent Tutoring System for Programming Education” for consideration for publication in *Applied Sciences*.

This paper presents CodeTutor-ITS, an intelligent tutoring system for programming education powered by the open-source Qwen2.5-7B-Instruct model. The system integrates four modules—dialogue tutoring, adaptive hints, exercise generation, and knowledge tracking—into a single locally-deployable platform. Unlike commercial LLM-based solutions, CodeTutor-ITS runs entirely on local hardware, ensuring zero operational cost and complete data privacy.

We believe this work is well-suited for *Applied Sciences* because it addresses a practical need in programming education with a rigorous experimental evaluation. The system is evaluated through five experiments covering response quality, prompt strategy comparison, module ablation, supplementary metrics, and fine-tuning analysis. All claims are supported by statistical tests with effect sizes reported, and cross-family validation using LLaMA-3-70B mitigates same-family bias. The complete system is publicly available on GitHub for reproducibility.

The manuscript has not been published elsewhere and is not under consideration by another journal. All authors have read and approved the manuscript and agree with its submission to *Applied Sciences*. The authors declare no conflicts of interest.

During the preparation of this manuscript, the authors used large language models for text editing and grammar checking. The authors have reviewed and edited the output and take full responsibility for the content of this publication.

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Thank you for considering our manuscript. We look forward to your response.

Sincerely,

Huanjie Yu
Corresponding Author, on behalf of all authors